

## Laplace Transform, Units and Initial Conditions

### 1 Question

We developed a differential equation to describe the surge velocity of a USV as

$$M\dot{v}(t) + Dv(t) = f(t)$$

where  $M$  is constant mass in [kg],  $v(t)$  is the surge velocity in [m/s],  $D$  is linear drag in [N/(m/s)] and  $f(t)$  is an input force in [N]. We can solve this first-order ODE using Laplace transforms. If we take the Laplace transform of both sides we get

$$M[sV(s) - v(0)] + DV(s) = F(s)$$

The question is, are the units in the various terms in the above equation, including the initial condition  $v(0)$ , consistent.

### 2 Answer

First we need to look at the units of the Laplace transform. Taking a function  $y(t)$  with units of “amplitude” [A], the Laplace transform is

$$Y(s) = \int_0^{\infty} y(t)e^{-st} dt$$

The variable “s” is the *frequency variable* often written as  $s = \sigma \pm j\omega$  with units of 1/seconds or [1/s] (sorry for all the s’s). Time has units of [s]. If  $y(t)$  has units of [A], then  $Y(s)$  has units of amplitude-per-frequency, [As] or [A/(rad/s)].

Returning to our original ODE, with units

$$M\dot{v}(t) + Dv(t) = f(t) \tag{1}$$

$$[kg][m/s^2] + [N/(m/s)][m/s] = [N] \tag{2}$$

where each term has units of [kg][m/s<sup>2</sup>]=[N]. In the Laplace domain

$$M[sV(s) - v(0)] + DV(s) = F(s) \tag{3}$$

$$MsV(s) - Mv(0) + DV(s) = F(s) \tag{4}$$

$$[kg][1/s][m] - [kg][m/s] + [N/(m/s)][m] = [Ns] \tag{5}$$

where each term has units of [kg][m/s]=[Ns]=[N/(rad/s)]